🎓 GW University RAG Chatbot

A Retrieval-Augmented Generation chatbot built on GW University bulletin PDFs. Ask anything about programs, tuition, admissions, and academic policies — answered with grounded citations from official GW documents.

STREAMLIT | GROQ API | LLAMA 3.3 70B | RAG PIPELINE

Live Demo

The chatbot loads GW University bulletin PDFs as its knowledge base and answers questions with source attribution — powered by LLaMA 3.3 70B via Groq.

> 👉 Try the live demo — no setup required. The knowledge base and API key are pre-configured.

Answers include document source tags showing which GW PDF was referenced and the similarity score for each retrieved chunk.

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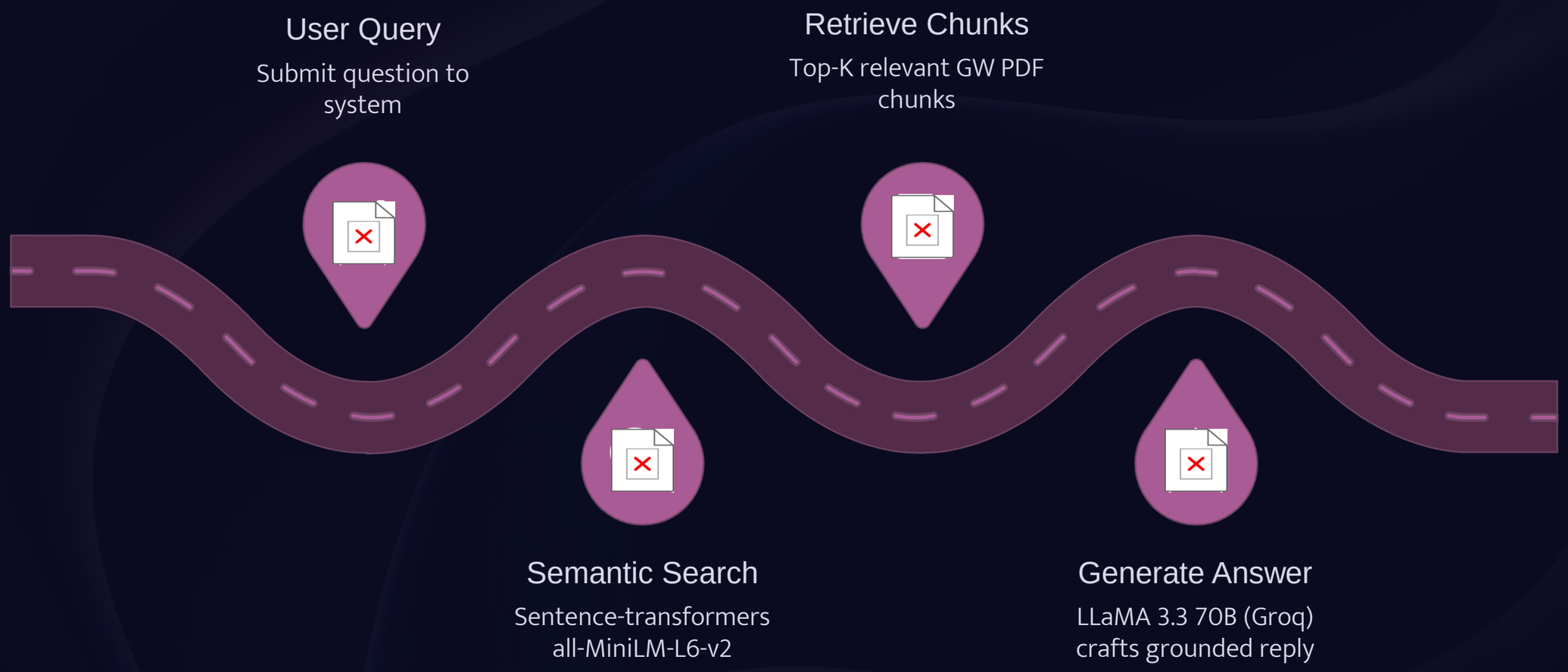
Architecture



Each question triggers a semantic search over pre-computed embeddings, retrieves the most relevant document chunks, and passes them as context to LLaMA 3.3 70B — which generates a concise, cited answer grounded in GW's official documents.

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🛠️ Tech Stack

Component	Technology
UI Framework	Streamlit
PDF Parsing	PyPDF2
Embedding Model	sentence-transformers (all-MiniLM-L6-v2)
Vector Search	NumPy cosine similarity
LLM	LLaMA 3.3 70B via Groq API
Index Persistence	Python pickle
Deployment	Streamlit Community Cloud

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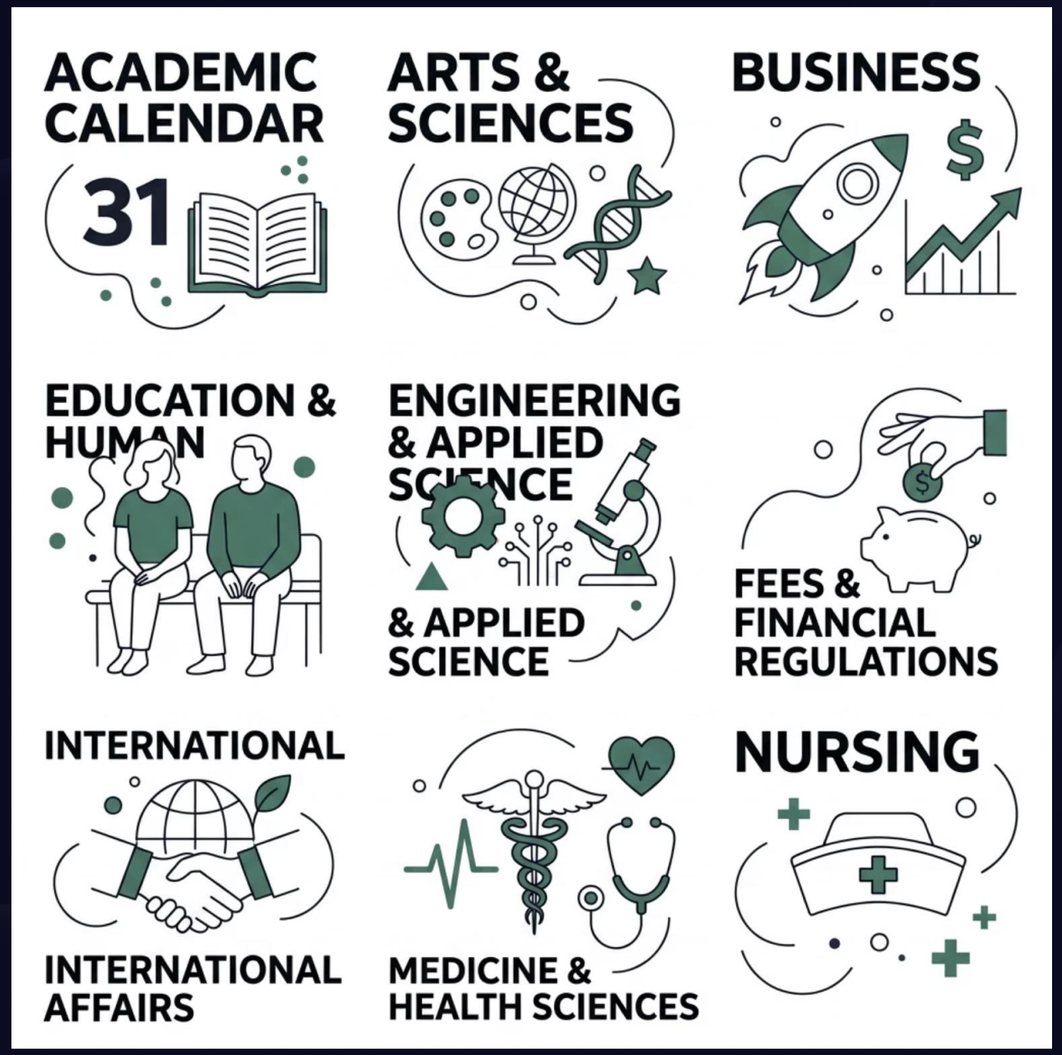
Knowledge Base

The chatbot is pre-indexed on GW University bulletin PDFs covering every school, college, and administrative policy. Each document is chunked, embedded, and persisted across sessions.

📌 Index persists in `gw_index.pkl` — no re-processing needed after the first build.



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Sample Questions

Admissions & GPA

What is the minimum GPA required for graduate students at GW?

Tuition & Fees

How much is full-time undergraduate tuition?

Grades & Policies

What happens if I receive an Incomplete grade?

Programs & Degrees

What engineering master's programs does GW offer?

Leaves & Transfers

How do I apply for a leave of absence? How do I transfer credits?

Academic Integrity

What is GW's academic integrity policy?

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🚀 Run Locally

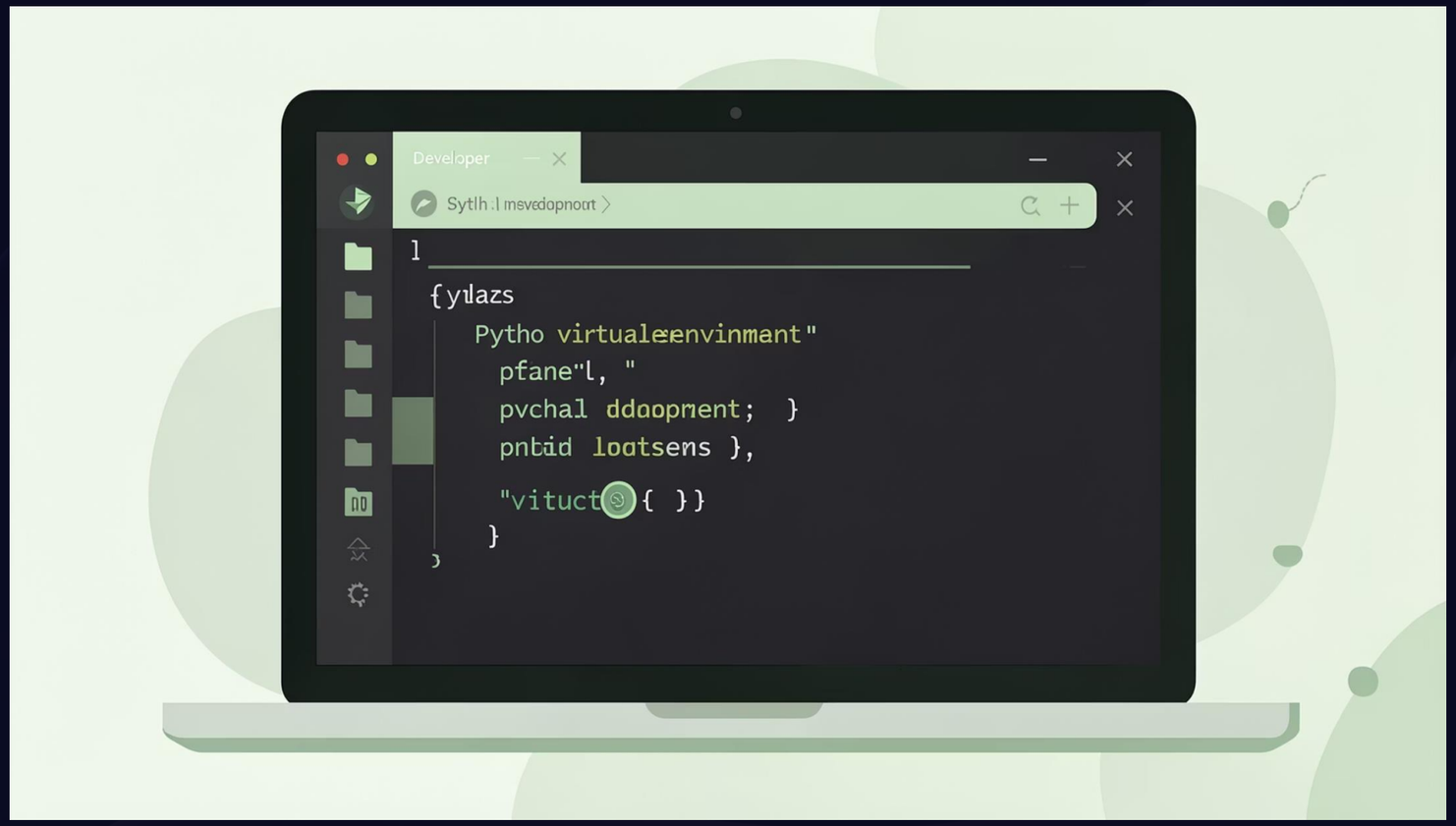
Prerequisites

- Python 3.10, 3.11, or 3.12 (**not** 3.14)
- Free Groq API key from console.groq.com

Steps

1. Clone the repo and `cd` into it
2. Create and activate a virtual environment
3. Run `pip install -r requirements.txt`
4. Run `streamlit run gw_chatbot_app_deployed.py`

App opens at `localhost:8501`. Enter your Groq API key, upload GW PDFs, and start chatting.



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Deploy on Streamlit Cloud

Push to GitHub
Fork or push your repo

New App
Go to share.streamlit.io and click New App

Configure
Set main file and add GROQ_API_KEY secret

Deploy
Click Deploy — app is live

Follow these four steps to deploy your GW RAG chatbot to Streamlit Community Cloud in under five minutes.

Secrets Configuration

Add your Groq API key under **Advanced Settings → Secrets**:

```
GROQ_API_KEY = "your_groq_api_key_here"
```

Why Streamlit Cloud?

- Free hosting for public apps
- Automatic redeploys on git push
- Secrets managed securely, never in code
- Zero server maintenance required

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How It Works

01

PDF Ingestion

GW bulletin PDFs are parsed with PyPDF2 and split into overlapping chunks (250 words, 50-word overlap).

02

Embedding

Each chunk is embedded using `all-MiniLM-L6-v2` from sentence-transformers.

03

Index Persistence

Embeddings and chunks are saved to `gw_index.pkl`, loading instantly on restart.

04

Retrieval

Cosine similarity finds the top-K relevant chunks above a configurable similarity threshold.

05

Generation

Retrieved context is passed to LLaMA 3.3 70B via Groq, which generates a grounded, cited answer.

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Key Takeaways

🎯 Grounded Answers

Every response is anchored in official GW bulletin PDFs — no hallucinations, full source attribution.

⚡ Fast & Free

LLaMA 3.3 70B via Groq delivers high-quality generation at inference speeds ideal for interactive chat.

🔄 Persistent Index

The pickle-based index survives app restarts — build once, query instantly every session.

🌐 One-Click Deploy

Streamlit Community Cloud + Secrets makes sharing the chatbot with your team effortless.